

# YINUO HAN

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## ACADEMIC APPOINTMENTS

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1. **Barr Fellow, Caltech, USA** Since Oct 2024  
Postdoctoral fellowship, Planetary Sciences, Division of Geological and Planetary Sciences

## EDUCATION

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1. **PhD in Astronomy, University of Cambridge, UK** Oct 2020 - Jul 2024  
Institute of Astronomy · Trinity College · Gates Cambridge Scholar  
**Thesis:** *The Structure of Extrasolar Planetesimal Belts in Images*  
**Advisor:** Prof. Mark Wyatt
2. **BSc (Adv) (Hons), University of Sydney, Australia** Feb 2016 - Dec 2019  
Majors in Physics, Neuroscience · First Class Honours (Physics) and the University Medal  
**Thesis:** *A Dance with Dragons: The Enigmatic Wolf-Rayet Binary Apep*  
**Advisor:** Prof. Peter Tuthill
3. **Study Abroad at University of California, Berkeley, USA** Aug 2017 - Dec 2017

## RESEARCH INTERESTS

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- Debris disks, planetary system formation and evolution
- Wolf-Rayet stars, colliding-wind binaries, carbon dust production
- High-resolution imaging, optical/infrared imaging, interferometry

## ADVISING

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List of theses and research projects advised.

- *VLT observations of  $\beta$  Pic and proper motion constraints of its dust clump.* Since Jun 2025  
Summer research project (Caltech SURF) by Polaris Hayes.
- *ALMA imaging of WR112 and dust modeling with JWST/MIRI.* Since Jun 2025  
Summer research project (Caltech SURF) by Donglin Wu.
- *Joint ALMA and JWST/NIRCam modeling of the vertical structure of the AU Mic debris disk.* Since Jun 2025  
Summer research project (Caltech SURF) by Junyi Zhao.
- *Secular perturbation as the origin of the dust and gas clump of Beta Pictoris.* Oct 2022 – Jun 2023  
Cambridge Part III/Masters project by Faran Sarwar (co-advised with Mark Wyatt).
- *The structure of the Beta Pictoris debris disk inferred from multi-wavelength mid-infrared images.* Jul – Aug 2022  
Summer research project by Andrew Zhang.

## TEACHING

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1. **Topics in Astrophysics** (Distributions, Timescales, Tides, Planet Formation) 2023  
Institute of Astronomy, University of Cambridge  
Regular supervisions/tutorial classes for Part II Astrophysics/final-year undergraduate astrophysics.
2. **Structure and Evolution of Stars** 2022  
Institute of Astronomy, University of Cambridge  
Regular supervisions/tutorial classes for Part II Astrophysics/final-year undergraduate astrophysics.
3. **Oscillating Systems, Waves and Quantum Waves, Fields** 2021  
Trinity College, University of Cambridge  
Weekly supervisions/tutorial classes for Part IA Physics/first-year undergraduate physics.
4. **Physics 1B** (Electromagnetism, Fluids, Quantum Mechanics) 2019  
School of Physics, University of Sydney  
Weekly tutorial classes for first-year undergraduate physics.

## AWARDS

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### UNIVERSITY OF CAMBRIDGE

- Gates Cambridge Scholarship* 2020  
*Cambridge International Scholarship*, accepted Gates above instead 2020

### UNIVERSITY OF SYDNEY

- University Medal* 2019  
*Henry Chamberlain Russell Prize for Astronomy* for best astronomy Honours thesis 2019  
*University of Sydney Honours Scholarship* 2019  
*Faculty of Science Olympiad Scholarship* 2016 - 2019  
*Deas-Thomson Scholarship in Physics* for top of class 2018  
*W.I.B. Smith Prize in Experimental Physics* 2018  
*Dean's List of Academic Excellence* 2016, 2017, 2018  
*Sydney Scholars Award* 2016 - 2018  
*Denison Summer Research Scholarship* 2016, 2017  
*Sydney Abroad International Exchange Scholarship* 2017  
*University of Sydney Academic Merit Prize* 2016  
*Levey Scholarship in Physics* for top of class 2016  
*Smith Prize in Experimental Physics* 2016

### OTHER

- University of Oxford Clarendon Scholarship*, declined 2020  
*International Physics Olympiad Bronze Medal* 2015

## TALKS

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### RESEARCH TALKS

- Conference: Exploring Star–Planet–Disk Connections *Oct 2025*  
*High-resolution ALMA imaging of extrasolar planetesimal belts.* *Tartu, Estonia*

- Seminar: MPIA Planet and Star Formation Seminar Oct 2025  
*The formation and evolution of dust in the colliding-wind binary Apep revealed by JWST.* Heidelberg, Germany
- Seminar: Columbia Astronomy & Astrophysics Seminar Mar 2025  
*Resolving radial substructures in debris disks.* New York City, USA
- Seminar: JPL Astrophysics Luncheon Seminar Feb 2025  
*Interpreting the Resolved Structure of Debris Disks.* Pasadena, USA
- Conference: Europlanet Science Congress 2024 Sep 2024  
*The high-resolution structure of debris disks revealed by the ARKS ALMA program.* Berlin, Germany
- Conference: Raising the veil on star formation near and far Apr 2024  
*High-resolution ALMA imaging of debris disks with the ARKS program.* Cambridge, UK
- Conference: Dust Devils: Debris Disks in the Sonoran Desert Mar 2024  
*The radial structure of debris disks in the ARKS ALMA program.* Tucson, USA
- Seminar: Department of Applied Mathematics and Theoretical Physics, University of Cambridge Oct 2023  
*The radial and vertical structure of debris disks.* Cambridge, UK
- Conference: Cosmic Dust 2023 Aug 2023  
*The physics of dusty Wolf-Rayet binaries revealed by observations of WR140.* Kitakyushu, Japan
- Seminar: Institute of Astronomy, University of Cambridge Oct 2022  
*Stellar fireworks: the spiral dust plumes of Wolf-Rayet binaries.* Cambridge, UK
- Conference: UK Exoplanet Community Meeting 2022 Sep 2022  
*Recovering the structure of edge-on debris disks non-parametrically.* Edinburgh, UK
- Conference: Debris Disks at Home and Abroad Sep 2022  
*Recovering the structure of edge-on debris disks non-parametrically.* Jena, Germany
- Seminar: Institute of Astronomy, University of Cambridge Mar 2022  
*A non-parametric method to recover the structure of edge-on debris disks.* Cambridge, UK

#### OUTREACH TALKS

- Institute of Astronomy Open Evening Oct 2023  
*What are exoplanets doing? Asteroid belts give us a clue.* Cambridge, UK
- Institute of Astronomy Open Evening Jan 2023  
*Stellar fireworks: a journey through the spirally dust of massive binary stars.* Cambridge, UK
- Integrated Science Class Seminar (Year 6–8), Ming Cheng School Apr 2022  
*Planets of the Solar System and beyond.* Beijing, China

- Gates Cambridge Teach-a-thon Feb 2021  
*Planets of the Solar System and beyond.* Online
- Australian Council of Undergraduate Research Oct 2019  
*Highly anisotropic stellar winds in recently-identified Wolf-Rayet binary.* Newcastle, Australia

#### POSTER PRESENTATIONS

- Conference: Gordon Research Conference on the Origins of Solar Systems Jun 2025  
*The radial structure of debris disks in the ARKS ALMA program.* South Hadley, USA
- Conference: Gordon Research Seminar + Conference on the Origins of Solar Systems Jun 2023  
*Modeling the structure of debris disks and the dust clump of Beta Pictoris.* South Hadley, USA
- Conference: STScI Spring Symposium: Planetary Systems and the Origins of Life in the Era of JWST May 2023  
*Modeling the structure of debris disks and the dust clump of Beta Pictoris.* Baltimore, USA
- Conference: Protostars and Planets VII Apr 2023  
*Has the dust clump in the debris disk of Beta Pictoris moved?* Kyoto, Japan

#### OBSERVING

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- *ALMA co-I. program:* Testing a new picture of debris disk gas by observing CI towards HD 121617 (ID 2025.1.00273.S, P.I. Gianni Cataldi) C.12, 2025
- *VLT/VISIR P.I. program:* Testing the giant impact scenario to explain the dust clump and cat's tail in Beta Pictoris (ID 114.27CQ) P114, 2024
- *ALMA P.I. program:* First resolved ALMA imaging of carbon-rich Wolf-Rayet binary dust structures (ID 2024.1.00803.S) C.11, 2024
- *ALMA co-I. program:* Resolving the extended structure associated with Apep's enigmatic wind (ID 2024.1.01244.S, P.I. Joseph Callingham) C.10, 2024
- *JWST P.I. program:* What lies beyond the inner spiral of Apep? (ID 5842) C.3, 2024
- *JWST P.I. program:* What causes warm dust interior to planetesimal belts? (ID 5709) C.3, 2024
- *VLT/VISIR P.I. program:* Wind kinematics in the extreme colliding-wind binary Apep: the first Galactic long-duration gamma-ray burst progenitor? (ID 113.26A7) P113, 2024
- *ALMA P.I. program:* First ALMA imaging of dust around Wolf-Rayet binaries (ID 2023.1.00999.S) C.10, 2023
- *ALMA co-I. program:* Vertical structure and planetary system dynamics (ID 2023.1.00487.S and 2025.1.00362.S, P.I. Meredith Hughes) C.10/12, 2023/2025
- *JWST co-I. program:* Fingerprinting the history of episodic dust creation in Wolf-Rayet binaries (ID 4093, P.I. Noel Richardson) C.2, 2023
- *Australia Telescope Compact Array:* Executing observations of Wolf-Rayet binary Apep 2019

## ADDITIONAL EXPERIENCE

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### REFEREE & PEER REVIEW

*Referee for MNRAS and AAS Journals*  
*ALMA distributed peer review, Cycles 8 – 12*  
*ESO distributed peer review*

### GRANTS

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|-----------------------------------|------|
| P.I., JWST Program 5842 (USD 78k) | 2024 |
| P.I., JWST Program 5709 (USD 74k) | 2024 |

### COLLABORATIONS

*ARKS: ALMA Large Program to image debris disks at high resolution, P.I. Sebastian Marino*

- *Leading Radial Structure team*

*JWST High Contrast Imaging: JWST ERS program imaging exoplanets and disks, P.I. Sasha Hinkley*

- *Nonparametric modeling of JWST 4-quadrant phase mask observations and application to HD 141569*

*DustERS: JWST ERS program on Wolf-Rayet binary dust formation, P.I. Ryan Lau*

- *Column density modeling of MIRI imaging and leading follow-up proposals*

### PRESS

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|--------------------------------------------------------------------------------------------|----------|
| <i>NASA, STScI and Caltech press releases based on Han et al. 2025.</i>                    | Nov 2025 |
| <i>University of Cambridge press release based on Han et al. 2022.</i>                     | Oct 2022 |
| Reported by major science and mainstream news outlets (e.g., ABC, BBC, CNN, The Guardian). |          |
| <i>University of Sydney press release based on Han et al. 2020.</i>                        | Oct 2020 |
| Reported by 110 news items from 21 countries in 8 languages (e.g., CNN).                   |          |

### EMPLOYMENT

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|---------------------------------------------------------------------------|-------------|
| <i>Supervisor (tutorial teaching), Institute of Astronomy, Cambridge</i>  | 2022 - 2023 |
| <i>Supervisor (tutorial teaching), Trinity College, Cambridge</i>         | 2020 - 2021 |
| <i>Tutor (tutorial teaching), School of Physics, University of Sydney</i> | 2019        |
| <i>Strategist, Goldman Sachs</i>                                          | 2018        |

### COMMUNITY INVOLVEMENT & OUTREACH

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|-----------------------------------------------------------------------------|-------------|
| <i>Public talk speaker, Institute of Astronomy, Cambridge open evenings</i> | 2022 - 2023 |
| <i>Technology Officer, Gates Cambridge Scholars Council</i>                 | 2020 - 2021 |
| <i>General Executive, Sydney University Mathematics Society</i>             | 2017 - 2019 |
| <i>Ambassador, Australian Science Innovations</i>                           | 2016 - 2019 |
| <i>Interviewer, Berkeley Scientific Journal</i>                             | 2017        |

## PUBLICATIONS

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### FIRST-AUTHOR PUBLICATIONS

7. [In press] **Y. Han**, E. Mansell, J. Jennings, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) II. The Radial Structure of Debris Disks. *Astronomy & Astrophysics* (2025).

6. **Y. Han**, R. M. T. White, J. R. Callingham, *et al.* The formation and evolution of dust in the colliding-wind Wolf-Rayet binary Apep observed by JWST. *The Astrophysical Journal* (2025). doi.org/10.3847/1538-4357/ae12e5
5. **Y. Han**, M. C. Wyatt, S. Marino. Recovering the structure of debris disks non-parametrically from images. *Monthly Notices of the Royal Astronomical Society* (2025). doi.org/10.1093/mnras/staf282
4. **Y. Han**, M. C. Wyatt, W. R. F. Dent. Has the dust clump in the debris disk of Beta Pictoris moved? *Monthly Notices of the Royal Astronomical Society* (2023). doi.org/10.1093/mnras/stac3769
3. **Y. Han**, P. G. Tuthill, R. M. Lau, A. Soulain. Radiation-driven acceleration in the expanding WR140 dust shell. *Nature* (2022). doi.org/10.1038/s41586-022-05155-5
2. **Y. Han**, M. C. Wyatt, L. Matrà. RAVE: a non-parametric method for recovering the surface brightness and height profiles of edge-on debris disks. *Monthly Notices of the Royal Astronomical Society* (2022). doi.org/10.1093/mnras/stac373
1. **Y. Han**, P. G. Tuthill, A. Soulain, J. R. Callingham, P. M. Williams, P. A. Crowther, B. J. S. Pope, B. Marcote. The extreme colliding-wind system Apep: resolved imagery of the central binary and dust plume in the infrared. *Monthly Notices of the Royal Astronomical Society* (2020). doi.org/10.1093/mnras/staa2349

#### IN PREPARATION AND IN REVIEW

2. [*In prep.*] **Y. Han**, M. C. Wyatt, K. Y. L. Su, *et al.* A radially broad collisional cascade in the debris disk of  $\gamma$  Oph revealed by JWST.
1. [*In prep.*] **Y. Han**, M. C. Wyatt, M. Jankovic, *et al.* The vertical structure of the Beta Pictoris debris disk viewed at mid-infrared and mm wavelengths.

#### CO-AUTHORED PUBLICATIONS

20. [*In press*] S. Marino, L. Matrà, A. M. Hughes, J. Ehrhardt, G. M. Kennedy, C. del Burgo, A. Brennan, **Y. Han**, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS). I: Motivation, Sample, Data Reduction and Results Overview. *Astronomy & Astrophysics* (2025).
19. [*In press*] B. Zawadzki, A. Fehr, A. M. Hughes, E. Mansell, J. Kittling, **Y. Han**, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS). III: The Vertical Structure of Debris Discs. *Astronomy & Astrophysics* (2025).
18. [*In press*] S. Mac Manamon, L. Matrà, S. Marino, A. Brennan, **Y. Han**, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS). IV: CO gas imaging and overview. *Astronomy & Astrophysics* (2025).
17. [*In press*] J. B. Lovell, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS). VI: Asymmetries and Offsets. *Astronomy & Astrophysics* (2025).
16. [*In press*] P. Weber, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS). IX: Gas-driven origin for the continuum arc in the debris disc of HD121617. *Astronomy & Astrophysics* (2025).

15. [In press] M. Jankovic, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS). X: Interpreting the peculiar dust rings around HD 131835. *Astronomy & Astrophysics* (2025).
14. R. M. T. White, B. J. S. Pope, P. G. Tuthill, **Y. Han**, *et al.* The Serpent Eating Its Own Tail: Dust Destruction in the Apep Colliding-Wind Nebula. *The Astrophysical Journal* (2025). doi.org/10.3847/1538-4357/adfbel
13. J. D. Monnier, **Y. Han**, *et al.* Revealing the Accelerating Wind in the Inner Region of Colliding-wind Binary WR 112. *The Astrophysical Journal* (2025). doi.org/10.3847/1538-3881/adfa03
12. N. D. Richardson, *et al.* Carbon-rich dust injected into the interstellar medium by Galactic WC binaries survives for hundreds of years. *The Astrophysical Journal* (2025). 10.3847/1538-4357/addf30
11. M. Sommer, M. C. Wyatt, **Y. Han**. A PR drag origin for the Fomalhaut disk’s pervasive inner dust: constraints on collisional strengths, icy composition, and embedded planets. *Monthly Notices of the Royal Astronomical Society* (2025). doi.org/10.1093/mnras/staf494
10. E. P. Lieb, *et al.* Dynamic imprints of colliding-wind dust formation from WR140. *The Astrophysical Journal Letters* (2025). doi.org/10.3847/2041-8213/ad9aa9
9. R. M. Lau, *et al.* A first look with JWST aperture masking interferometry (AMI): resolving circumstellar dust around the Wolf-Rayet binary WR 137 beyond the Rayleigh limit. *The Astrophysical Journal* (2024). doi.org/10.3847/2041-8213/ad9aa9
8. J. Terrill, S. Marino, R. A. Booth, **Y. Han**, J. Jennings, M. C. Wyatt. Deprojecting and constraining the vertical thickness of exoKuiper belts. *Monthly Notices of the Royal Astronomical Society* (2023). doi.org/10.1093/mnras/stad1847
7. R. M. Lau, *et al.* From dust to nanodust: resolving circumstellar dust from the colliding-wind binary Wolf-Rayet 140. *The Astrophysical Journal* (2022). doi.org/10.3847/1538-4357/acd4c5
6. R. M. Lau, M. J. Hankins, **Y. Han**, *et al.* Nested dust shells around the Wolf-Rayet binary WR 140 observed with JWST. *Nature Astronomy* (2022). doi.org/10.1038/s41550-022-01812-x
5. P. A. Robinson, X. Gao, **Y. Han**. Relationships between lognormal distributions of neural properties, activity, criticality, and connectivity. *Biological Cybernetics* (2021). doi.org/10.1007/s00422-021-00871-z
4. B. Marcote, J. R. Callingham, M. De Becker, P. G. Edwards, **Y. Han**, R. Schulz, J. Stevens, P. G. Tuthill. AU-scale radio imaging of the wind collision region in the brightest and most luminous non-thermal colliding wind binary Apep. *Monthly Notices of the Royal Astronomical Society* (2021). doi.org/10.1093/mnras/staa3863
3. R. M. Lau, M. J. Hankins, **Y. Han** *et al.* Resolving periodic spirals and shadows from the Wolf-Rayet dust factory WR112. *The Astrophysical Journal* (2020). doi.org/10.3847/1538-4357/abaab8
2. J. R. Callingham, P. A. Crowther, P. M. Williams, P. G. Tuthill, **Y. Han**, B. J. S. Pope. Two Wolf-Rayet stars at the heart of colliding-wind binary Apep. *Monthly Notices of the Royal Astronomical Society* (2020). doi.org/10.1093/mnras/staa1244

1. M. Li, **Y. Han**, M. J. Aburn, M. Breakspear, R. Q. Poldrack, J. M. Shine, J. T. Lizier. Transitions in information processing dynamics at the whole-brain network level are driven by alterations in neural gain. *PLOS Computational Biology* (2019). doi.org/10.1371/journal.pcbi.1006957